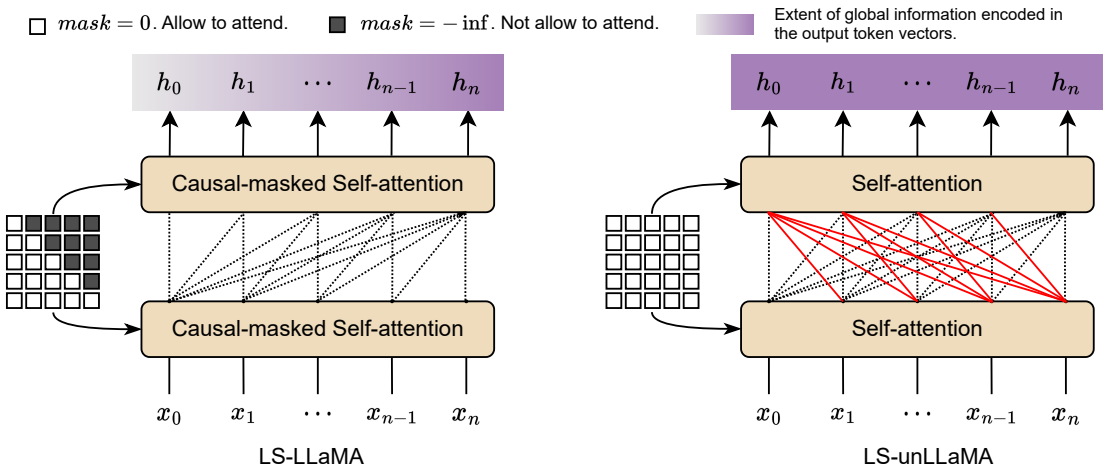


Figure 2: An illustration of the effects caused by different masking methods in LS-LLaMA and LS-unLLaMA.

### 2.2 Label-supervised unmasked LLaMA

The causal masks $CM$, as shown in Equation 2,

$$CM = \begin{bmatrix} 0 & -\inf & -\inf & \cdots & -\inf & -\inf & -\inf \\ 0 & 0 & -\inf & \cdots & -\inf & -\inf & -\inf \\ 0 & 0 & 0 & \cdots & -\inf & -\inf & -\inf \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & -\inf & -\inf \\ 0 & 0 & 0 & \cdots & 0 & 0 & -\inf \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}, \tag{2}$$

in decoder blocks prevent information leaking, as the decoder is only allowed to attend to earlier positions in text generation. Bidirectional dependency extraction of the self-attention layer is reduced to single-direction, leading to critical information loss at the token level. Our empirical studies show that using token representations learned with causal masks significantly underperforms in token classification tasks. To address such an issue, we remove the causal masks from `LlamaForTokenClassification`[5] and extract the latent representations $H'$ for token classification, as illustrated in Figure 2,

$$H'_{tkn} = \text{unmasked_LlmaForTokenClf}(T). \tag{3}$$

The essential bidirectional information is expected to be replenished in token representations during finetuning as all the tokens can attend to each other.

We believe the essential global dependency is also helpful in the sequence classification, and hence, remove the causal masks in `LlamaForSequenceClassification`. With bidirectional self-attention resumed, we have more choices in pooling. We have tested three pooling methods, i.e., `max`, `average`, and `last`, and the experiments show that max-over-time pooling yields better performance than average pooling and last-token pooling in classification tasks without causal masks.

## 3 Experiment

### 3.1 Tasks and evaluation metrics

We have conducted extensive experiments on text classification and NER tasks against zero- and few-shot LLMs, instruction-following LLMs, and discriminant baselines to validate the effectiveness of the proposed label-supervised adaptation method based on LLaMA-2-7B.

[5]Note that this operation is different from the padding mask and requires modification on LLaMA's source code.

#### 3.1.1 Multiclass text classification

Experiments were conducted on four English datasets from multiple domains, i.e., SST2 and SST5[6] (general sentiment analysis), AGNews (news categorization), and Twitter Financial News[7] (short as "Twitter-Fin", financial sentiment analysis). Table 1 contains the statistics and label classes of the datasets.

Moreover, we experimented on the German, English, Spanish, and Chinese subsets of Multilingual Amazon Reviews Corpus[8] [Keung et al., 2020] to test LS-LLaMA and LS-unLLaMA under the multilingual setting. The datasets contain Amazon product reviews for 31 product categories. The task is to predict the product category given the review content. For each language, the subset contains 200,000, 5,000, and 5,000 data samples in the training, development, and test sets respectively. We report the accuracy of text classification tasks.

Table 1: Dataset descriptions for multiclass text classification.

| Dataset | Train | Test | Classes |
|---|---|---|---|
| SST2 | 67,300 | 872* | "Positive", "Negative" |
| SST5 | 8,540 | 2,210 | "Very negative", "Negative", "Neutral", "Positive", "Very positive" |
| AGNews | 120,000 | 7,600 | "World", "Sports", "Business', "Sci/Tech" |
| Twitter Fin | 9,938 | 2,486 | "Bearish", "Bullish", "Neutral" |

* Testing on the validation set for all baselines

#### 3.1.2 Named entity recognition

For the NER task, we experiment on OntoNotes V5.0[9] and CoNLL2003 [Tjong Kim Sang and De Meulder, 2003], which are widely-adopted datasets for information retrieval. OntoNotes V5.0 has 59,924 training samples and 8,262 testing samples, and CoNLL2003 contains 14,987 training samples and 3,684 testing samples. We report the F1 scores on the NER tasks.

### 3.2 Implementation details

LLaMA-2-chat under the zero-shot setting was tested on the classification tasks. LLaMA-2-chat-7B and LLaMA-2-chat-13B were deployed on Nvidia GeForce RTX4090 and A100 GPUs, respectively.

The main experiments of label-supervised adaptation are based on LLaMA-2-7B. These experiments were run on a single Nvidia GeForce RTX4090 GPU. We adopt the standard parameter-efficient fine-tuning method LoRA to finetune the model. We configured the LoRA settings *lora_rank*, *lora_alpha*, and *lora_dropout* to 12, 32, and 0.1, respectively. We set the batch size to 8 and initial learning rate to $8e-5$ using grid search. For the sequence classification, we use the *last token* pooling on LS-LLaMA and the *max* pooling on LS-unLLaMA according to the ablation study on the pooling method.

We apply instruction-tuning based on LLaMA-2-7B on text classification tasks. The prompt is set as follows:

> `prompt_input` = "You are a classification model. Based on the given article, you need to predict the most relevant category label from `{all_labels}`. One article has only one label. \n ### Input article: `{input_text}` \n ### Output: "

The instruction-following model is finetuned for 50k steps on large datasets (training samples exceed 50,000) and 10k steps on small datasets (training samples less than 50,000) with a batch size of 16 and an initial learning rate of $2e-4$.

### 3.3 Experimental results

#### 3.3.1 Multiclass text classification

We present the results of multiclass text classification experiments in Table 2, offering a comprehensive analysis of LLMs under different adaptation settings and discriminant models. Notably, LLMs such as LLaMA-2-7B/13B and GPT-3-175B exhibit suboptimal performance when subjected to zero- and few-shot settings. While instruction-tuning

[6] `https://huggingface.co/datasets/SetFit/sst5`
[7] `https://huggingface.co/datasets/zeroshot/twitter-financial-news-sentiment`
[8] `https://huggingface.co/datasets/amazon_reviews_multi`
[9] `https://catalog.ldc.upenn.edu/LDC2013T19`

Table 2: Accuracy (%) in multiclass text classification.

| Models | SST2 | AGNews | Twitter-Fin | SST5 |
|---|---|---|---|---|
| *Zero- and few-shot* | | | | |
| LLaMA-2-7B (zero-shot) | 76.26 | 37.39 | 23.40 | 39.05 |
| LLaMA-2-13B (zero-shot) | 69.90 | 59.40 | 38.74 | 37.01 |
| GPT-3 175B (zero-shot) | 54.3† | 43.9† | – | – |
| GPT-3 175B (few-shot) | 93.4† | 84.3† | – | – |
| *Instruction-tuning* | | | | |
| LLaMA-2-7B (instruction-tuning) | 91.97 | 52.40 | 68.72 | 43.35 |
| *Discriminant baselines* | | | | |
| BERT-Base | 92.78 | 94.51 | 88.19 | 55.07 |
| BERT-Large | 92.86 | 94.45 | 88.74 | 55.79 |
| RoBERTa-Base | 94.61 | 94.70 | 90.32 | 58.46 |
| RoBERTa-Large | 96.10 | 94.78 | 90.95 | 59.64 |
| AGN [Li et al., 2021] | 93.27 | 93.82 | – | 55.72 |
| *Ours* | | | | |
| LS-LLaMA-2-7B | 96.67 | 95.38 | **91.87** | **62.31** |
| LS-LLaMA-2-13B | 96.90 | 95.66 | 91.20 | 62.17 |
| LS-unLLaMA-2-7B | **97.36** | **95.68** | 91.54 | 60.50 |
| LS-unLLaMA-2-13B | 92.77 | 95.44 | 87.94 | 52.99 |

† results reported in Zhao et al. [2021]

brings about notable enhancements on LLaMA-2-7B, it is apparent that the achieved results are not comparable to the discriminative capabilities demonstrated by BERT and RoBERTa. This observation highlights the challenges faced by conversational LLMs when predicting from a predefined label set.

Our proposed label-supervised adaptation method showcases substantial improvements over instruction fine-tuning. Notably, the magnitude of these enhancements is more outstanding in datasets where labels are more domain-specific and demand a deeper understanding of commonsense knowledge. For instance, in SST2, which features commonly-used binary sentiment labels ("Positive" and "Negative"), we observe an absolute improvement of 5.39% (equivalent to a relative improvement of 5.86%). In contrast, in SST5 with fine-grained sentiment labels like "Neutral", "Very positive", and "Very negative", our approach demonstrates an 18.96% absolute improvement (equivalent to a relative improvement of 43.74%). When applied to the Twitter-Fin dataset, which comprises domain-specific financial labels ("Bullish" and "Bearish"), our method produces a 23.15% absolute improvement (equivalent to a relative improvement of 33.69%). In the AGNews dataset, labeled with domain-specific news categories, we achieve a 43.28% absolute improvement (equivalent to an astonishing relative improvement of 82.60%). Remarkably, such improvements do not require sophisticated prompt engineering and external knowledge.

Furthermore, it is noteworthy that both LS-LLaMA and LS-unLLaMA consistently demonstrate significant improvements when compared to robust discriminative baselines such as BERT-Large and RoBERTa-Large. Our approach achieves improvements of 1.26%, 0.90%, 0.92%, and 2.67% when compared against RoBERTa-Large on SST2, AGNews, Twitter-Fin, and SST5, respectively. Interestingly, we also observe variations in performance between LS-LLaMA and LS-unLLaMA. Specifically, LS-unLLaMA outperforms LS-LLaMA on SST2 and AGNews, but exhibits inferior performance on Twitter-Fin and SST5. A potential explanation for these differences can be drawn from the dataset sizes, as indicated in Table 1. SST2 and AGNews are notably larger datasets compared to Twitter-Fin and SST5, a distinction that may provide insights. Since we removed causal masks from LS-unLLaMA, the model needs more training samples to reconstruct the parameters that were previously concealed during pretraining. This process is not strictly needed in LS-LLaMA as it can directly employ the last token for classification task. Therefore, we conclude that LS-LLaMA can quickly adapt on small-scale datasets, and LS-unLLaMA can achieve even better results with ample training samples.

### 3.3.2 Multilingual multiclass text classification

According to Touvron et al. [2023b], LLaMA-2's pretraining data primarily consists of English text, accounting for a substantial 89.70%, with non-English languages representing only a minority within the pretraining dataset—specifically, 0.17% German, 0.13% Chinese, and 0.13% Spanish. However, "[a] training corpus with a majority in English means